

Milestone_2_Movie_Data_Acquisition

May 29, 2026

1 Milestone 2: Movie Data Acquisition

Cleaning/Formatting Flat File Source

Perform at least 5 data transformation and/or cleansing steps to your flat file data. The below examples are not required - they are just potential transformations you could do. If your data doesn't work for these scenarios, complete different transformations. You can do the same transformation multiple times if needed to clean your data. The goal is a clean dataset at the end of the milestone.

Examples:

- Replace Headers.
- Format data into a more readable format.
- Identify outliers and bad data.
- Find duplicates.
- Fix casing or inconsistent values.
- Conduct Fuzzy Matching.

Make sure you clearly number and label each transformation step (Step #1, Step #2, etc.) in your code and describe what it is doing in 1-2 sentences.

```
[1]: # Load the necessary libraries.  
import pandas as pd  
import numpy as np  
from scipy import stats
```

1.1 I'm working with two different .csv files from the DataSet I mentioned on Milestone 1.

```
[2]: # Load .csv files containing critic reviews and movies information into pandas  
      ↪DataFrames for analysis.  
critic_reviews_df = pd.read_csv("rotten_tomatoes_critic_reviews.csv")  
movies_df = pd.read_csv("rotten_tomatoes_movies.csv")
```

2 Critics Review DataFrame.

```
[3]: # Display first few rows of critic reviews DataFrame.  
critic_reviews_df.head()
```

```
[3]: rotten_tomatoes_link    critic_name  top_critic    publisher_name \
0      m/0814255  Andrew L. Urban    False    Urban Cinefile
1      m/0814255    Louise Keller    False    Urban Cinefile
2      m/0814255          NaN    False    FILMINK (Australia)
3      m/0814255    Ben McEachen    False    Sunday Mail (Australia)
4      m/0814255    Ethan Alter    True    Hollywood Reporter

    review_type review_score review_date \
0      Fresh          NaN  2010-02-06
1      Fresh          NaN  2010-02-06
2      Fresh          NaN  2010-02-09
3      Fresh    3.5/5  2010-02-09
4      Rotten          NaN  2010-02-10

                                review_content
0  A fantasy adventure that fuses Greek mythology...
1  Uma Thurman as Medusa, the gorgon with a coiff...
2  With a top-notch cast and dazzling special eff...
3  Whether audiences will get behind The Lightnin...
4  What's really lacking in The Lightning Thief i...
```

```
[4]: # Get the original dimensions of the critic reviews DataFrame.
critic_reviews_df.shape
```

```
[4]: (1130017, 8)
```

```
[5]: # Display the data types of each column in the DataFrame to understand the
      ↪structure of the data.
critic_reviews_df.dtypes
```

```
[5]: rotten_tomatoes_link    object
critic_name                object
top_critic                 bool
publisher_name             object
review_type               object
review_score              object
review_date               object
review_content            object
dtype: object
```

```
[6]: # STEP 1.
# Create a new DataFrame cleaned_critic_reviews_df from critic_reviews_df
cleaned_critic_reviews_df = critic_reviews_df.copy()
```

```
[7]: # STEP 2.
# Format Headers: Removing underscores and capitalize the first letter of each
      ↪word.
```

```
cleaned_critic_reviews_df.columns = cleaned_critic_reviews_df.columns.str.  
    ↪replace("_", " ").str.title()
```

```
[8]: # Display first few rows of cleaned_critic_reviews_df to verify change to  
    ↪headers.  
cleaned_critic_reviews_df.head()
```

```
[8]: Rotten Tomatoes Link      Critic Name  Top Critic      Publisher Name \  
0      m/0814255  Andrew L. Urban      False      Urban Cinefile  
1      m/0814255  Louise Keller      False      Urban Cinefile  
2      m/0814255      NaN      False      FILMINK (Australia)  
3      m/0814255  Ben McEachen      False      Sunday Mail (Australia)  
4      m/0814255  Ethan Alter      True      Hollywood Reporter
```

```
Review Type Review Score Review Date \  
0      Fresh      NaN  2010-02-06  
1      Fresh      NaN  2010-02-06  
2      Fresh      NaN  2010-02-09  
3      Fresh      3.5/5  2010-02-09  
4      Rotten      NaN  2010-02-10
```

```
Review Content  
0  A fantasy adventure that fuses Greek mythology...  
1  Uma Thurman as Medusa, the gorgon with a coiff...  
2  With a top-notch cast and dazzling special eff...  
3  Whether audiences will get behind The Lightnin...  
4  What's really lacking in The Lightning Thief i...
```

```
[9]: # Calculate the total number of missing values in each column of the  
    ↪cleaned_critic_reviews_df.  
missing_values = cleaned_critic_reviews_df.isnull().sum()  
print(missing_values)
```

```
Rotten Tomatoes Link      0  
Critic Name      18529  
Top Critic      0  
Publisher Name      0  
Review Type      0  
Review Score      305936  
Review Date      0  
Review Content      65806  
dtype: int64
```

```
[10]: # STEP 3.  
# Fill missing values in "Critic Name" with the placeholder "Unknown Critic."  
cleaned_critic_reviews_df["Critic Name"] = cleaned_critic_reviews_df["Critic_  
    ↪Name"].fillna("Unknown Critic")
```

```
[11]: # STEP 4.
# Define a function to clean the Review Scores and create a new column named
# Numeric Review Score.
def clean_review_score(value):
    if pd.isna(value) or value == "":
        return np.nan # Keep empty cells as NaN.
    elif "/" in value:
        return float(value.split("/")[0]) # Extract numeric value before "/".
    elif isinstance(value, str) and value.replace("-", "").isalpha(): # Check
    # for grades.
        return np.nan # Treat grades as NaN.
    else:
        return np.nan # Default case for unexpected formats.

# Apply "clean_review_score" function to "Review Score" column and store the
# cleaned numeric values in the "Numeric Review Score" column.
cleaned_critic_reviews_df["Numeric Review Score"] =
cleaned_critic_reviews_df["Review Score"].apply(clean_review_score)

# Calculate the mean score, ignoring NaN values.
mean_review_score = cleaned_critic_reviews_df["Numeric Review Score"].mean()

# Fill NaN values in the new column with the mean score
cleaned_critic_reviews_df["Numeric Review Score"] =
cleaned_critic_reviews_df["Numeric Review Score"].fillna(mean_review_score)

# Display "Mean Review Score".
print("Mean Review Score: ", mean_review_score)
```

Mean Review Score: 3.9343797641805165

```
[12]: # STEP 5.
# Fill missing values in Review Content with N/A.
cleaned_critic_reviews_df["Review Content"] = cleaned_critic_reviews_df["Review
Content"].fillna("N/A")
```

```
[13]: # Display the total number of missing values in each column of the DataFrame
# after cleaning.
# "Numeric Review Score" column is a cleaned version of "Review Score" column.
missing_values = cleaned_critic_reviews_df.isnull().sum()
print(missing_values)
```

Rotten Tomatoes Link	0
Critic Name	0
Top Critic	0
Publisher Name	0
Review Type	0
Review Score	305936

```

Review Date          0
Review Content        0
Numeric Review Score  0
dtype: int64

```

```

[14]: # STEP 6.
      # Convert "Review Date" from string/object to datetime format.
      cleaned_critic_reviews_df["Review Date"] = pd.
      ↪to_datetime(cleaned_critic_reviews_df["Review Date"], errors = "coerce")

```

```

[15]: # Check unique values from the Review Type column to identify distinct review_
      ↪categories.
      unique_values = cleaned_critic_reviews_df["Review Type"].unique()
      print("Unique values in Review Type: ", unique_values)

```

```

Unique values in Review Type:  ['Fresh' 'Rotten']

```

```

[16]: # STEP 7.
      # Check for duplicate reviews based on "Rotten Tomatoes Link" and "Critic Name."
      duplicate_reviews = cleaned_critic_reviews_df.duplicated(subset = ["Rotten_
      ↪Tomatoes Link", "Critic Name"], keep = False)

      # # Display duplicate rows, if there are any.
      duplicate_entries = cleaned_critic_reviews_df[duplicate_reviews]
      print(duplicate_entries)

      # Remove duplicates, keeping the first occurrence.
      cleaned_critic_reviews_df.drop_duplicates(subset = ["Rotten Tomatoes Link",
      ↪"Critic Name"], inplace = True)

```

	Rotten Tomatoes Link	Critic Name	Top Critic	\
2	m/0814255	Unknown Critic	False	
80	m/0814255	Unknown Critic	False	
101	m/0814255	Unknown Critic	True	
320	m/1000013-12_angry_men	Unknown Critic	False	
348	m/1000013-12_angry_men	Unknown Critic	False	
...	...	...	...	
1129695	m/zootopia	Jonathan Sánchez	False	
1129696	m/zootopia	Jonathan Sánchez	False	
1129987	m/zulu	Unknown Critic	False	
1130005	m/zulu	Unknown Critic	False	
1130006	m/zulu	Unknown Critic	False	

	Publisher Name	Review Type	Review Score	Review Date	\
2	FILMINK (Australia)	Fresh	NaN	2010-02-09	
80	National Post	Rotten	2.5/4	2010-02-12	
101	Time Out	Rotten	2/5	2010-02-16	
320	TV Guide	Fresh	4/5	2001-11-11	
348	Film4	Fresh	NaN	2009-04-24	

...	...	...	...	...
1129695	Time Out México	Fresh	4/5	2016-02-19
1129696	Cinema Movil	Fresh	4/5	2016-02-19
1129987	TV Guide	Fresh	5/5	2000-01-01
1130005	Sky Cinema	Fresh	5/5	2017-07-10
1130006	Empire Magazine	Fresh	4/5	2017-07-10

	Review Content \
2	With a top-notch cast and dazzling special eff...
80	N/A
101	It's not enough to compensate for the computer...
320	Cinema heresy that worked.
348	Not much action, lots of words, great cinema.
...	...
1129695	Zootopia has all the elements to become a clas...
1129696	This is probably one of the best films Disney ...
1129987	This amazing film is devastatingly accurate in...
1130005	This is that rarity in films - an all-action, ...
1130006	As a spectacular war film with a powerful mora...

	Numeric Review Score
2	3.93438
80	2.50000
101	2.00000
320	4.00000
348	3.93438
...	...
1129695	4.00000
1129696	4.00000
1129987	5.00000
1130005	5.00000
1130006	4.00000

[259268 rows x 9 columns]

```
[17]: # Get the shape of cleaned_critic_reviews_df.
cleaned_critic_reviews_df.shape
```

[17]: (996470, 9)

```
[18]: # Display cleaned_critic_reviews_df types of each column in the DataFrame.
cleaned_critic_reviews_df.dtypes
```

```
[18]: Rotten Tomatoes Link      object
Critic Name                  object
Top Critic                   bool
Publisher Name               object
Review Type                  object
```

```

Review Score          object
Review Date           datetime64[ns]
Review Content         object
Numeric Review Score   float64
dtype: object

```

```

[19]: # Display first few rows of the cleaned_critic_reviews_df DataFrame.
cleaned_critic_reviews_df.head()

```

```

[19]:  Rotten Tomatoes Link      Critic Name  Top Critic      Publisher Name \
0      m/0814255  Andrew L. Urban      False      Urban Cinefile
1      m/0814255  Louise Keller      False      Urban Cinefile
2      m/0814255  Unknown Critic      False      FILMINK (Australia)
3      m/0814255  Ben McEachen      False      Sunday Mail (Australia)
4      m/0814255  Ethan Alter      True      Hollywood Reporter

```

```

      Review Type Review Score Review Date \
0      Fresh      NaN  2010-02-06
1      Fresh      NaN  2010-02-06
2      Fresh      NaN  2010-02-09
3      Fresh      3.5/5  2010-02-09
4      Rotten      NaN  2010-02-10

```

```

      Review Content  Numeric Review Score
0  A fantasy adventure that fuses Greek mythology...      3.93438
1  Uma Thurman as Medusa, the gorgon with a coiff...      3.93438
2  With a top-notch cast and dazzling special eff...      3.93438
3  Whether audiences will get behind The Lightnin...      3.50000
4  What's really lacking in The Lightning Thief i...      3.93438

```

```

[20]: # Display the cleaned DataFrame to review the changes made during the cleaning
      ↪process.
print(cleaned_critic_reviews_df)

```

```

      Rotten Tomatoes Link      Critic Name  Top Critic \
0      m/0814255  Andrew L. Urban      False
1      m/0814255  Louise Keller      False
2      m/0814255  Unknown Critic      False
3      m/0814255  Ben McEachen      False
4      m/0814255  Ethan Alter      True
...
1130012  m/zulu_dawn  Chuck O'Leary      False
1130013  m/zulu_dawn      Ken Hanke      False
1130014  m/zulu_dawn  Dennis Schwartz      False
1130015  m/zulu_dawn  Christopher Lloyd      False
1130016  m/zulu_dawn  Brent McKnight      False

```

```

      Publisher Name Review Type Review Score Review Date \

```

0	Urban Cinefile	Fresh	NaN	2010-02-06
1	Urban Cinefile	Fresh	NaN	2010-02-06
2	FILMINK (Australia)	Fresh	NaN	2010-02-09
3	Sunday Mail (Australia)	Fresh	3.5/5	2010-02-09
4	Hollywood Reporter	Rotten	NaN	2010-02-10
...	...	...	...	...
1130012	Fantastica Daily	Rotten	2/5	2005-11-02
1130013	Mountain Xpress (Asheville, NC)	Fresh	3.5/5	2007-03-07
1130014	Dennis Schwartz Movie Reviews	Fresh	B+	2010-09-16
1130015	Sarasota Herald-Tribune	Rotten	3.5/5	2011-02-28
1130016	The Last Thing I See	Rotten	C	2020-07-09

Review Content \

0	A fantasy adventure that fuses Greek mythology...
1	Uma Thurman as Medusa, the gorgon with a coiff...
2	With a top-notch cast and dazzling special eff...
3	Whether audiences will get behind The Lightnin...
4	What's really lacking in The Lightning Thief i...
...	...
1130012	N/A
1130013	Seen today, it's not only a startling indictme...
1130014	A rousing visual spectacle that's a prequel of...
1130015	A simple two-act story: Prelude to war, and th...
1130016	Rides the line between being a pure artifact o...

Numeric Review Score

0	3.93438
1	3.93438
2	3.93438
3	3.50000
4	3.93438
...	...
1130012	2.00000
1130013	3.50000
1130014	3.93438
1130015	3.50000
1130016	3.93438

[996470 rows x 9 columns]

```
[21]: # Calculate the Z-scores for 'Numeric Review Score.'
cleaned_critic_reviews_df["z_score"] = stats.
↳ zscore(cleaned_critic_reviews_df["Numeric Review Score"])

# Identify outliers (e.g., Z-score > 3 or < -3).
outliers = cleaned_critic_reviews_df[(cleaned_critic_reviews_df["z_score"] > 3) |
↳ (cleaned_critic_reviews_df["z_score"] < -3)]
```



```
# Display the outliers.
print(outliers)
```

	Rotten Tomatoes Link	Critic Name	Top Critic	\
262	m/0878835	Philip Martin	False	
319	m/1000013-12_angry_men	Brian Webster	False	
349	m/1000013-12_angry_men	Dan Jardine	False	
607	m/1000123-310_to_yuma	Brian Webster	False	
662	m/1000224-accused	Jamie Gillies	False	
...	...	...	...	
1129317	m/zoolander	Mike DeWolfe	False	
1129573	m/zoolander_2	Piers Marchant	False	
1129906	m/zootopia	Philip Martin	False	
1129907	m/zootopia	Dan Lybarger	False	
1129983	m/zorba_the_greek	Dan Jardine	False	

	Publisher Name	Review Type	Review Score	Review Date	\
262	Arkansas Democrat-Gazette	Rotten	83/100	2010-07-03	
319	Apollo Guide	Fresh	92/100	2001-04-18	
349	Apollo Guide	Fresh	82/100	2009-08-03	
607	Apollo Guide	Fresh	76/100	2007-08-27	
662	Apollo Guide	Fresh	87/100	2002-04-30	
...	...	...	...	...	
1129317	Apollo Guide	Fresh	75/100	2001-10-04	
1129573	Arkansas Democrat-Gazette	Rotten	75/100	2016-11-15	
1129906	Arkansas Democrat-Gazette	Fresh	88/100	2016-08-26	
1129907	Arkansas Democrat-Gazette	Fresh	86/100	2016-11-14	
1129983	Cinemania	Fresh	79/100	2011-08-20	

	Review Content	\
262	It is well-observed but morally stunted, in th...	
319	While the composition of juries might change o...	
349	taut, well-crafted sociological study	
607	An enjoyable western that's an excellent remin...	
662	N/A	
...	...	
1129317	If you have ever watched a fashion model strut...	
1129573	Whatever inspiration first fed the original fi...	
1129906	...an Animal Farm-type fable about racism and ...	
1129907	The [DMV} sequence is so funny that it's oddly...	
1129983	The serious critique in Zorba attends the trag...	

	Numeric Review Score	z_score
262	83.0	14.104411
319	92.0	15.709865
349	82.0	13.926028
607	76.0	12.855725

662	87.0	14.817946
...	...	...
1129317	75.0	12.677341
1129573	75.0	12.677341
1129906	88.0	14.996330
1129907	86.0	14.639562
1129983	79.0	13.390876

[6005 rows x 10 columns]

- A total of 6005 rows were analyzed, and outliers were identified as those with Z-scores greater than 3 or less than -3. These include scores like 83.0, 92.0, and higher, indicating they are exceptionally high.
- The outliers suggest unusually positive reviews and may represent specific critics or contexts contributing to these high scores.
- Significant outliers in the review scores, warrant further investigation to understand their implications. ns.

```
[22]: # Check for unexpected values in categorical columns.
print(cleaned_critic_reviews_df["Critic Name"].value_counts())
print(cleaned_critic_reviews_df["Review Type"].value_counts())
print(cleaned_critic_reviews_df["Review Content"].value_counts())
```

```
Critic Name
Unknown Critic      7121
Emanuel Levy        7031
Dennis Schwartz     5821
Roger Ebert         5701
Brian Orndorf       5415
...
Joseph P. Kahn      1
Spencer Doar        1
Peter Momtchiloff   1
Henry Giardina      1
Will Heaven         1
Name: count, Length: 11109, dtype: int64
Review Type
Fresh      634968
Rotten     361502
Name: count, dtype: int64
Review Content
N/A
53352
Parental Content Review
245
full review at Movies for the Masses
180
full review in Greek
```

92

.

37

...

Much like the WWE, is damn fine entertainment, not to be taken too seriously, and full of laughs, cheers and some surprisingly heartfelt moments.

1

Like the characters it celebrates, this knockabout British comedy is a little rough around the edges, but its heart is in the right place.

1

Fighting with My Family is a film with a great heart especially from its lead, Florence Pugh. [Full Review in Spanish]

1

A buoyant, sunny movie, driven by performances from a cast unafraid of charming vulgarity.

1

Rides the line between being a pure artifact of a bygone era-one that depicts a romanticized, unrealistic version of the time-and a modern, revisionist take that strives for a more accurate portrayal of historical reality. 1

Name: count, Length: 940543, dtype: int64

Critic Name: The dataset shows a significant concentration of reviews from a few critics, with “Unknown Critic” contributing **7121 reviews**. This indicates potential bias and over-reliance on certain voices.

Review Type: A notable predominance of “Fresh” reviews **634968** compared to “Rotten” reviews **361502** suggests a generally positive bias in the dataset.

Review Content: Many entries are marked as N/A **53352**, indicating missing review content. This raises concerns about data completeness. Other entries vary widely, with some unique reviews only appearing once, pointing to a diverse range of opinions but limited utility due to high N/A counts.

The analysis highlights potential biases in critic representation and a positive skew in review types while also identifying significant gaps in the review content that could affect further analyses.

```
[23]: # Check if 'Review Date' has any invalid dates or outliers
print(cleaned_critic_reviews_df["Review Date"].min(),
      ↪cleaned_critic_reviews_df["Review Date"].max())
```

1800-01-01 00:00:00 2020-10-29 00:00:00

- The dataset shows a review date range from **January 1, 1800** to **October 29, 2020**.
- The early date of 1800 is likely an invalid entry, as it predates modern film criticism and raises concerns about data accuracy.
- This suggests potential data entry errors or inconsistencies that need to be addressed.

3 Movies DataFrame.

```
[24]: movies_df.head()
```

```
[24]:
```

	rotten_tomatoes_link \	movie_title \	movie_info \	critics_consensus content_rating \	genres directors \	authors \	actors original_release_date \
0	m/0814255	Percy Jackson & the Olympians: The Lightning T...	Always trouble-prone, the life of teenager Per...	Though it may seem like just another Harry Pot... PG	Action & Adventure, Comedy, Drama, Science Fic... Chris Columbus	Craig Titley, Chris Columbus, Rick Riordan	
1	m/0878835	Please Give	Kate (Catherine Keener) and her husband Alex (...)	Nicole Holofcener's newest might seem slight i... R	Comedy Nicole Holofcener	Nicole Holofcener	
2	m/10	10	A successful, middle-aged Hollywood songwriter...	Blake Edwards' bawdy comedy may not score a pe... R	Comedy, Romance Blake Edwards	Blake Edwards	
3	m/1000013-12_angry_men	12 Angry Men (Twelve Angry Men)	Following the closing arguments in a murder tr...	Sidney Lumet's feature debut is a superbly wri... NR	Classics, Drama Sidney Lumet	Reginald Rose	
4	m/1000079-20000_leagues_under_the_sea	20,000 Leagues Under The Sea	In 1866, Professor Pierre M. Aronnax (Paul Luk...	One of Disney's finest live-action adventures,... G	Action & Adventure, Drama, Kids & Family Richard Fleischer	Earl Felton	

0	Logan Lerman, Brandon T. Jackson, Alexandra Da...	2010-02-12
1	Catherine Keener, Amanda Peet, Oliver Platt, R...	2010-04-30
2	Dudley Moore, Bo Derek, Julie Andrews, Robert ...	1979-10-05
3	Martin Balsam, John Fiedler, Lee J. Cobb, E.G...	1957-04-13
4	James Mason, Kirk Douglas, Paul Lukas, Peter L...	1954-01-01

	production_company	tomatometer_status	tomatometer_rating	\
0	20th Century Fox	Rotten	49.0	
1	Sony Pictures Classics	Certified-Fresh	87.0	
2	Waner Bros.	Fresh	67.0	
3	Criterion Collection	Certified-Fresh	100.0	
4	Disney	Fresh	89.0	

	tomatometer_count	audience_status	audience_rating	audience_count	\
0	149.0	Spilled	53.0	254421.0	
1	142.0	Upright	64.0	11574.0	
2	24.0	Spilled	53.0	14684.0	
3	54.0	Upright	97.0	105386.0	
4	27.0	Upright	74.0	68918.0	

	tomatometer_top_critics_count	tomatometer_fresh_critics_count	\
0	43	73	
1	44	123	
2	2	16	
3	6	54	
4	5	24	

	tomatometer_rotten_critics_count
0	76
1	19
2	8
3	0
4	3

[5 rows x 22 columns]

```
[25]: # Get the original dimensions of the movies DataFrame.
movies_df.shape
```

```
[25]: (17712, 22)
```

```
[26]: # Display the data types of each column in the DataFrame to understand the
      ↪ structure of the data.
movies_df.dtypes
```

```
[26]: rotten_tomatoes_link    object
      movie_title            object
```

```

movie_info                object
critics_consensus         object
content_rating            object
genres                   object
directors                 object
authors                  object
actors                   object
original_release_date     object
streaming_release_date    object
runtime                   float64
production_company        object
tomatometer_status        object
tomatometer_rating        float64
tomatometer_count         float64
audience_status          object
audience_rating          float64
audience_count           float64
tomatometer_top_critics_count  int64
tomatometer_fresh_critics_count int64
tomatometer_rotten_critics_count int64
dtype: object

```

```

[27]: # STEP 1.
      # Create a new DataFrame cleaned_movies_df from movies_df.
      cleaned_movies_df = movies_df.copy()

```

```

[28]: # STEP 2.
      # Format Headers: Removing underscores and capitalize the first letter of each
      ↪word.
      cleaned_movies_df.columns = cleaned_movies_df.columns.str.replace("_", " ").str.
      ↪title()

```

```

[29]: # Display first few rows of cleaned_critic_reviews_df to verify change to
      ↪headers.
      cleaned_movies_df.head()

```

```

[29]:          Rotten Tomatoes Link \
0                m/0814255
1                m/0878835
2                m/10
3      m/1000013-12_angry_men
4  m/1000079-20000_leagues_under_the_sea

          Movie Title \
0  Percy Jackson & the Olympians: The Lightning T...
1                Please Give
2                10

```

3 12 Angry Men (Twelve Angry Men)
 4 20,000 Leagues Under The Sea

Movie Info \

0 Always trouble-prone, the life of teenager Per...
 1 Kate (Catherine Keener) and her husband Alex (...
 2 A successful, middle-aged Hollywood songwriter...
 3 Following the closing arguments in a murder tr...
 4 In 1866, Professor Pierre M. Aronnax (Paul Luk...

Critics Consensus Content Rating \

0 Though it may seem like just another Harry Pot... PG
 1 Nicole Holofcener's newest might seem slight i... R
 2 Blake Edwards' bawdy comedy may not score a pe... R
 3 Sidney Lumet's feature debut is a superbly wri... NR
 4 One of Disney's finest live-action adventures,... G

Genres Directors \

0 Action & Adventure, Comedy, Drama, Science Fic... Chris Columbus
 1 Comedy Nicole Holofcener
 2 Comedy, Romance Blake Edwards
 3 Classics, Drama Sidney Lumet
 4 Action & Adventure, Drama, Kids & Family Richard Fleischer

Authors \

0 Craig Titley, Chris Columbus, Rick Riordan
 1 Nicole Holofcener
 2 Blake Edwards
 3 Reginald Rose
 4 Earl Felton

Actors Original Release Date \

0 Logan Lerman, Brandon T. Jackson, Alexandra Da... 2010-02-12
 1 Catherine Keener, Amanda Peet, Oliver Platt, R... 2010-04-30
 2 Dudley Moore, Bo Derek, Julie Andrews, Robert ... 1979-10-05
 3 Martin Balsam, John Fiedler, Lee J. Cobb, E.G... 1957-04-13
 4 James Mason, Kirk Douglas, Paul Lukas, Peter L... 1954-01-01

	Production Company	Tomatometer	Status	Tomatometer	Rating \
0	20th Century Fox		Rotten	49.0	
1	Sony Pictures Classics	Certified-Fresh		87.0	
2	Waner Bros.	Fresh		67.0	
3	Criterion Collection	Certified-Fresh		100.0	
4	Disney	Fresh		89.0	

	Tomatometer Count	Audience Status	Audience Rating	Audience Count \
0	149.0	Spilled	53.0	254421.0

1	142.0	Upright	64.0	11574.0
2	24.0	Spilled	53.0	14684.0
3	54.0	Upright	97.0	105386.0
4	27.0	Upright	74.0	68918.0

	Tomatometer Top Critics Count	Tomatometer Fresh Critics Count	\
0	43		73
1	44		123
2	2		16
3	6		54
4	5		24

	Tomatometer Rotten Critics Count
0	76
1	19
2	8
3	0
4	3

[5 rows x 22 columns]

```
[30]: # Calculate the total number of missing values in each column of the
      ↪ "cleaned_movies_df."
      movies_missing_values = cleaned_movies_df.isnull().sum()
      print(movies_missing_values)
```

Rotten Tomatoes Link	0
Movie Title	0
Movie Info	321
Critics Consensus	8578
Content Rating	0
Genres	19
Directors	194
Authors	1542
Actors	352
Original Release Date	1166
Streaming Release Date	384
Runtime	314
Production Company	499
Tomatometer Status	44
Tomatometer Rating	44
Tomatometer Count	44
Audience Status	448
Audience Rating	296
Audience Count	297
Tomatometer Top Critics Count	0
Tomatometer Fresh Critics Count	0
Tomatometer Rotten Critics Count	0

dtype: int64

```
[31]: # STEPS 3 - 11.
# Fill missing values in categorical (object) columns with the appropriate
# placeholder.
cleaned_movies_df["Movie Info"] = cleaned_movies_df["Movie Info"].
    .fillna("Unknown Info")
cleaned_movies_df["Critics Consensus"] = cleaned_movies_df["Critics Consensus"].
    .fillna("No Consensus")
cleaned_movies_df["Genres"] = cleaned_movies_df["Genres"].fillna("Unknown
    Genre")
cleaned_movies_df["Directors"] = cleaned_movies_df["Directors"].fillna("Unknown
    Director")
cleaned_movies_df["Authors"] = cleaned_movies_df["Authors"].fillna("Unknown
    Author")
cleaned_movies_df["Actors"] = cleaned_movies_df["Actors"].fillna("Unknown
    Actors")
cleaned_movies_df["Production Company"] = cleaned_movies_df["Production
    Company"].fillna("Unknown Production Company")
cleaned_movies_df["Tomatometer Status"] = cleaned_movies_df["Tomatometer
    Status"].fillna("Unknown Status")
cleaned_movies_df["Audience Status"] = cleaned_movies_df["Audience Status"].
    .fillna("Unknown Audience Status")
```

```
[32]: # STEPS 12 - 15.
# Fill missing numerical (float64) columns with the median.
cleaned_movies_df["Tomatometer Rating"] = cleaned_movies_df["Tomatometer
    Rating"].fillna(cleaned_movies_df["Tomatometer Rating"].median())
cleaned_movies_df["Tomatometer Count"] = cleaned_movies_df["Tomatometer Count"].
    .fillna(cleaned_movies_df["Tomatometer Count"].median())
cleaned_movies_df["Audience Rating"] = cleaned_movies_df["Audience Rating"].
    .fillna(cleaned_movies_df["Audience Rating"].median())
cleaned_movies_df["Audience Count"] = cleaned_movies_df["Audience Count"].
    .fillna(cleaned_movies_df["Audience Count"].median())
```

```
[33]: # STEPS 16 - 17.
# Convert "Original Release Date" and "Streaming Release Date" to datetime
# format
cleaned_movies_df["Original Release Date"] = pd.
    .to_datetime(cleaned_movies_df["Original Release Date"], errors = "coerce")
cleaned_movies_df["Streaming Release Date"] = pd.
    .to_datetime(cleaned_movies_df["Streaming Release Date"], errors = "coerce")
```

```
[34]: # STEPS 18 - 19.
# Fill missing "Original Release Date" and "Streaming Release Date" with a
# placeholder date
```

```
cleaned_movies_df["Original Release Date"] = cleaned_movies_df["Original_
↳Release Date"].fillna(pd.to_datetime("1800-01-01"))
cleaned_movies_df["Streaming Release Date"] = cleaned_movies_df["Streaming_
↳Release Date"].fillna(pd.to_datetime("1800-01-01"))
```

```
[35]: # Step 20.
# Fill missing values in 'Runtime' with 0 as a placeholder.
cleaned_movies_df["Runtime"] = cleaned_movies_df["Runtime"].fillna(0)
```

```
[36]: # Recheck missing values in each column of the "cleaned_movies_df."
movies_missing_values = cleaned_movies_df.isnull().sum()
print(movies_missing_values)
```

```
Rotten Tomatoes Link      0
Movie Title                0
Movie Info                0
Critics Consensus         0
Content Rating            0
Genres                    0
Directors                 0
Authors                   0
Actors                    0
Original Release Date     0
Streaming Release Date    0
Runtime                   0
Production Company        0
Tomatometer Status        0
Tomatometer Rating        0
Tomatometer Count         0
Audience Status          0
Audience Rating          0
Audience Count           0
Tomatometer Top Critics Count 0
Tomatometer Fresh Critics Count 0
Tomatometer Rotten Critics Count 0
dtype: int64
```

```
[37]: # Check unique values 'Tomatometer Status' and 'Audience Status' to identify_
↳distinct review categories.
print(cleaned_movies_df['Tomatometer Status'].unique())
print(cleaned_movies_df['Audience Status'].unique())
```

```
['Rotten' 'Certified-Fresh' 'Fresh' 'Unknown Status']
['Spilled' 'Upright' 'Unknown Audience Status']
```

```
[38]: # Check for duplicates based on "Rotten Tomatoes Link"
duplicates_rotten_tomatoes = cleaned_movies_df.duplicated(subset = "Rotten_
↳Tomatoes Link", keep = False)
```

```
# Check for duplicates based on "Movie Title".
duplicates_movie_title = cleaned_movies_df.duplicated(subset = "Movie Title",
↳keep = False)

# Display duplicate rows based on "Rotten Tomatoes Link" and "Movie Title"
print(cleaned_movies_df[duplicates_rotten_tomatoes])
print(cleaned_movies_df[duplicates_movie_title])
```

Empty DataFrame

Columns: [Rotten Tomatoes Link, Movie Title, Movie Info, Critics Consensus, Content Rating, Genres, Directors, Authors, Actors, Original Release Date, Streaming Release Date, Runtime, Production Company, Tomatometer Status, Tomatometer Rating, Tomatometer Count, Audience Status, Audience Rating, Audience Count, Tomatometer Top Critics Count, Tomatometer Fresh Critics Count, Tomatometer Rotten Critics Count]

Index: []

[0 rows x 22 columns]

	Rotten Tomatoes Link	Movie Title \
7	m/1000123-310_to_yuma	3:10 to Yuma
10	m/10002114-dark_water	Dark Water
17	m/10003276-criminal	Criminal
22	m/10003925-dead_end	Dead End
28	m/10004288-running_scared	Running Scared
...	...	...
17496	m/wonder_woman_2017	Wonder Woman
17553	m/wuthering_heights	Wuthering Heights
17554	m/wuthering_heights_2011	Wuthering Heights
17670	m/z	Z
17671	m/z_2019	Z

	Movie Info \
7	Dan Evans (Van Heflin), a drought-plagued Ariz...
10	In this moody Japanese horror film, newly-sing...
17	Needing a new partner capable of intricate con...
22	When a family en route to a Christmas Eve gath...
28	Mafia flunky Joey Gazelle (Paul Walker) is ord...
...	...
17496	Before she was Wonder Woman (Gal Gadot), she w...
17553	Young orphan Heathcliff (Ralph Fiennes) is ado...
17554	A poor English boy is taken in by a wealthy fa...
17670	Repression is the rule of the day in this film...
17671	Unknown Info

	Critics Consensus	Content Rating \
7	No Consensus	NR
10	No Consensus	PG-13

17	If you saw Nine Queens, it may feel redundant,...	R
22	No Consensus	R
28	This film runs with frenetic energy punctuated...	R
...	...	...
17496	Thrilling, earnest, and buoyed by Gal Gadot's ...	PG-13
17553	No Consensus	PG
17554	Director Andrea Arnold's gritty, naturalistic ...	NR
17670	Powerfully effective, this anti-fascist politi...	PG
17671	No Consensus	NR

	Genres \
7	Classics, Drama, Western
10	Art House & International, Horror, Mystery & S...
17	Drama, Mystery & Suspense
22	Comedy, Horror, Mystery & Suspense
28	Action & Adventure, Drama, Mystery & Suspense
...	...
17496	Action & Adventure, Drama, Science Fiction & F...
17553	Drama, Romance
17554	Drama, Romance
17670	Art House & International, Drama, Mystery & Su...
17671	Horror

	Directors \
7	Delmer Daves
10	Hideo Nakata
17	Greggory Jacobs, Gregory Jacobs, Greg Jacobs
22	Jean-Baptiste Andrea, Fabrice Canepa, Jean-Bap...
28	Wayne Kramer
...	...
17496	Patty Jenkins
17553	Peter Kosminsky
17554	Andrea Arnold
17670	Costa-Gavras
17671	Brandon Christensen

	Authors \
7	Halsted Welles
10	Hideo Hakata, Takashige Ichise, Yoshihiro Naka...
17	Gregory Jacobs, Fabián Bielinsky, Steven Soder...
22	Jean-Baptiste Andrea, Fabrice Canepa
28	Wayne Kramer
...	...
17496	Allan Heinberg, Geoff Johns
17553	Anne Devlin
17554	Andrea Arnold, Olivia Hetreed
17670	Jorge Semprún, Costa-Gavras
17671	Brandon Christensen, Colin Minihan

	Actors \
7	Glenn Ford, Van Heflin, Felicia Farr, Leora Da...
10	Hitomi Kuroki, Rio Kanno, Shigemitsu Ogi, Mire...
17	John C. Reilly, Diego Luna, Maggie Gyllenhaal,...
22	Ray Wise, Lin Shaye, Mick Cain, Alexandra Hold...
28	Paul Walker, Cameron Bright, Vera Farmiga, Kar...
...	...
17496	Gal Gadot, Chris Pine, Connie Nielsen, Robin W...
17553	Juliette Binoche, Ralph Fiennes, Janet McTeer,...
17554	James Howson, Kaya Scodelario, Steve Evets, Ni...
17670	Yves Montand, Irene Papas, Jean-Louis Trintign...
17671	Keegan Connor Tracy, Jett Klyne, Sean Rogerson...

	Original Release Date ...	Production Company \
7	1957-08-07 ...	Columbia Pictures
10	2002-01-19 ...	Toho Company Ltd
17	2004-09-24 ...	Warner Bros. Pictures
22	2003-09-26 ...	LionsGate Entertainment
28	2006-02-24 ...	New Line Cinema
...	...	...
17496	2017-06-02 ...	Warner Bros. Pictures
17553	1800-01-01 ...	Paramount Pictures
17554	2012-10-05 ...	Oscilloscope Pictures
17670	1969-02-26 ...	Rialto Pictures
17671	1800-01-01 ...	Digital Interference Productions

	Tomatometer Status	Tomatometer Rating	Tomatometer Count \
7	Fresh	96.0	28.0
10	Fresh	80.0	15.0
17	Fresh	69.0	124.0
22	Fresh	75.0	8.0
28	Rotten	41.0	131.0
...	...	...	...
17496	Certified-Fresh	93.0	458.0
17553	Rotten	25.0	12.0
17554	Fresh	68.0	113.0
17670	Certified-Fresh	93.0	46.0
17671	Unknown Status	67.0	28.0

	Audience Status	Audience Rating	Audience Count \
7	Upright	79.0	9243.0
10	Upright	66.0	21475.0
17	Spilled	57.0	6711.0
22	Upright	60.0	10373.0
28	Upright	79.0	135400.0
...	...	...	...
17496	Upright	86.0	131171.0

17553	Upright	66.0	9807.0
17554	Spilled	47.0	8750.0
17670	Upright	93.0	6526.0
17671	Unknown Audience Status	63.0	4277.0

	Tomatometer Top Critics Count	Tomatometer Fresh Critics Count	\
7	6	27	
10	3	12	
17	35	85	
22	1	6	
28	39	54	
...	...	...	
17496	59	427	
17553	3	3	
17554	35	77	
17670	14	43	
17671	2	26	

	Tomatometer Rotten Critics Count
7	1
10	3
17	39
22	2
28	77
...	...
17496	32
17553	9
17554	36
17670	3
17671	1

[1157 rows x 22 columns]

```
[39]: # Steps 21 - 22.
# Remove duplicates based on "Rotten Tomatoes Link" while keeping the first
      ↪ occurrence.
cleaned_movies_df.drop_duplicates(subset = "Rotten Tomatoes Link", keep =
      ↪ "first", inplace = True)

# Remove duplicates based on "Movie Title"
cleaned_movies_df.drop_duplicates(subset = "Movie Title", keep = "first",
      ↪ inplace = True)

[40]: # Check for remaining duplicates in 'rotten_tomatoes_link' and 'movie_title'
print(f"Remaining duplicates in 'Rotten Tomatoes Link': {cleaned_movies_df.
      ↪ duplicated(subset = 'Rotten Tomatoes Link').sum()}")
```

```
print(f"Remaining duplicates in 'Movie Title': {cleaned_movies_df.  
↳ duplicated(subset = 'Movie Title').sum()}")
```

Remaining duplicates in 'Rotten Tomatoes Link': 0

Remaining duplicates in 'Movie Title': 0

```
[41]: # Get the shape of cleaned_movies_df.  
cleaned_movies_df.shape
```

[41]: (17106, 22)

```
[42]: # Display cleaned_movies_df types of each column in the DataFrame.  
cleaned_movies_df.dtypes
```

```
[42]: Rotten Tomatoes Link      object  
Movie Title                  object  
Movie Info                   object  
Critics Consensus            object  
Content Rating               object  
Genres                       object  
Directors                    object  
Authors                      object  
Actors                       object  
Original Release Date        datetime64[ns]  
Streaming Release Date        datetime64[ns]  
Runtime                      float64  
Production Company            object  
Tomatometer Status            object  
Tomatometer Rating            float64  
Tomatometer Count             float64  
Audience Status              object  
Audience Rating              float64  
Audience Count               float64  
Tomatometer Top Critics Count int64  
Tomatometer Fresh Critics Count int64  
Tomatometer Rotten Critics Count int64  
dtype: object
```

```
[43]: # Display first few rows of the cleaned_movies_df.  
cleaned_movies_df.head()
```

```
[43]:      Rotten Tomatoes Link \  
0      m/0814255  
1      m/0878835  
2      m/10  
3      m/1000013-12_angry_men  
4      m/1000079-20000_leagues_under_the_sea
```

	Movie Title \
0	Percy Jackson & the Olympians: The Lightning T...
1	Please Give
2	10
3	12 Angry Men (Twelve Angry Men)
4	20,000 Leagues Under The Sea

	Movie Info \
0	Always trouble-prone, the life of teenager Per...
1	Kate (Catherine Keener) and her husband Alex (...)
2	A successful, middle-aged Hollywood songwriter...
3	Following the closing arguments in a murder tr...
4	In 1866, Professor Pierre M. Aronnax (Paul Luk...

	Critics Consensus	Content Rating \
0	Though it may seem like just another Harry Pot...	PG
1	Nicole Holofcener's newest might seem slight i...	R
2	Blake Edwards' bawdy comedy may not score a pe...	R
3	Sidney Lumet's feature debut is a superbly wri...	NR
4	One of Disney's finest live-action adventures,...	G

	Genres	Directors \
0	Action & Adventure, Comedy, Drama, Science Fic...	Chris Columbus
1	Comedy	Nicole Holofcener
2	Comedy, Romance	Blake Edwards
3	Classics, Drama	Sidney Lumet
4	Action & Adventure, Drama, Kids & Family	Richard Fleischer

	Authors \
0	Craig Titley, Chris Columbus, Rick Riordan
1	Nicole Holofcener
2	Blake Edwards
3	Reginald Rose
4	Earl Felton

	Actors	Original Release Date \
0	Logan Lerman, Brandon T. Jackson, Alexandra Da...	2010-02-12
1	Catherine Keener, Amanda Peet, Oliver Platt, R...	2010-04-30
2	Dudley Moore, Bo Derek, Julie Andrews, Robert ...	1979-10-05
3	Martin Balsam, John Fiedler, Lee J. Cobb, E.G...	1957-04-13
4	James Mason, Kirk Douglas, Paul Lukas, Peter L...	1954-01-01

	Production Company	Tomatometer Status	Tomatometer Rating \
0	20th Century Fox	Rotten	49.0
1	Sony Pictures Classics	Certified-Fresh	87.0
2	Waner Bros.	Fresh	67.0
3	Criterion Collection	Certified-Fresh	100.0

4	...	Disney	Fresh	89.0
---	-----	--------	-------	------

	Tomatometer	Count	Audience Status	Audience Rating	Audience Count	\
0		149.0	Spilled	53.0	254421.0	
1		142.0	Upright	64.0	11574.0	
2		24.0	Spilled	53.0	14684.0	
3		54.0	Upright	97.0	105386.0	
4		27.0	Upright	74.0	68918.0	

	Tomatometer	Top Critics	Count	Tomatometer	Fresh Critics	Count	\
0			43			73	
1			44			123	
2			2			16	
3			6			54	
4			5			24	

	Tomatometer	Rotten Critics	Count
0			76
1			19
2			8
3			0
4			3

[5 rows x 22 columns]

```
[44]: # Display the cleaned DataFrame to review the changes made during the cleaning
      ↪process.
      print(cleaned_movies_df)
```

	Rotten Tomatoes Link	\
0	m/0814255	
1	m/0878835	
2	m/10	
3	m/1000013-12_angry_men	
4	m/1000079-20000_leagues_under_the_sea	
...	...	
17707	m/zoot_suit	
17708	m/zootopia	
17709	m/zorba_the_greek	
17710	m/zulu	
17711	m/zulu_dawn	

	Movie Title	\
0	Percy Jackson & the Olympians: The Lightning T...	
1	Please Give	
2	10	
3	12 Angry Men (Twelve Angry Men)	
4	20,000 Leagues Under The Sea	

...	...
17707	Zoot Suit
17708	Zootopia
17709	Zorba the Greek
17710	Zulu
17711	Zulu Dawn

Movie Info \

0	Always trouble-prone, the life of teenager Per...
1	Kate (Catherine Keener) and her husband Alex (...)
2	A successful, middle-aged Hollywood songwriter...
3	Following the closing arguments in a murder tr...
4	In 1866, Professor Pierre M. Aronnax (Paul Luk...

...	...
17707	Mexican-American gangster Henry Reyna (Daniel ...)
17708	From the largest elephant to the smallest shre...
17709	Traveling to inspect an abandoned mine his fat...
17710	In 1879, the Zulu nation hands colonial Britis...
17711	Sir Henry Bartle Frere's (John Mills) vastly o...

Critics Consensus Content Rating \

0	Though it may seem like just another Harry Pot...	PG
1	Nicole Holofcener's newest might seem slight i...	R
2	Blake Edwards' bawdy comedy may not score a pe...	R
3	Sidney Lumet's feature debut is a superbly wri...	NR
4	One of Disney's finest live-action adventures,...	G

...	...	...
17707	No Consensus	R
17708	The brilliantly well-rounded Zootopia offers a...	PG
17709	No Consensus	NR
17710	Zulu patiently establishes a cast of colorful ...	PG
17711	No Consensus	PG

Genres \

0	Action & Adventure, Comedy, Drama, Science Fic...
1	Comedy
2	Comedy, Romance
3	Classics, Drama
4	Action & Adventure, Drama, Kids & Family

...	...
17707	Drama, Musical & Performing Arts
17708	Action & Adventure, Animation, Comedy
17709	Action & Adventure, Art House & International,...
17710	Classics, Drama
17711	Action & Adventure, Art House & International,...

Directors \

0	Chris Columbus
---	----------------

1	Nicole Holofcener
2	Blake Edwards
3	Sidney Lumet
4	Richard Fleischer
...	...
17707	Luis Valdez
17708	Byron Howard, Rich Moore, Jared Bush
17709	Unknown Director
17710	Cy Endfield, Cyril Endfield
17711	Douglas Hickox

	Authors \
0	Craig Titley, Chris Columbus, Rick Riordan
1	Nicole Holofcener
2	Blake Edwards
3	Reginald Rose
4	Earl Felton
...	...
17707	Luis Valdez
17708	Jared Bush, Phil Johnston
17709	Unknown Author
17710	Cy Endfield, John Prebble
17711	Cy Endfield, Anthony Storey

	Actors \
0	Logan Lerman, Brandon T. Jackson, Alexandra Da...
1	Catherine Keener, Amanda Peet, Oliver Platt, R...
2	Dudley Moore, Bo Derek, Julie Andrews, Robert ...
3	Martin Balsam, John Fiedler, Lee J. Cobb, E.G...
4	James Mason, Kirk Douglas, Paul Lukas, Peter L...
...	...
17707	Daniel Valdez, Edward James Olmos, Charles Aid...
17708	J.K. Simmons, Kristen Bell, Octavia Spencer, A...
17709	Anthony Quinn, Alan Bates, Irene Papas, Lila K...
17710	Stanley Baker, Jack Hawkins, Ulla Jacobsson, J...
17711	Burt Lancaster, Peter O'Toole, Simon Ward, Joh...

	Original Release Date	...	Production Company \
0	2010-02-12	...	20th Century Fox
1	2010-04-30	...	Sony Pictures Classics
2	1979-10-05	...	Waner Bros.
3	1957-04-13	...	Criterion Collection
4	1954-01-01	...	Disney
...	...	...	...
17707	1981-10-02	...	MCA Universal Home Video
17708	2016-03-04	...	Walt Disney Animation Studios
17709	1964-12-17	...	Fox
17710	1964-06-17	...	Paramount Pictures

17711 1979-12-14 ... Tango Entertainment

	Tomatometer Status	Tomatometer Rating	Tomatometer Count	\
0	Rotten	49.0	149.0	
1	Certified-Fresh	87.0	142.0	
2	Fresh	67.0	24.0	
3	Certified-Fresh	100.0	54.0	
4	Fresh	89.0	27.0	
...	...	...	...	
17707	Rotten	56.0	9.0	
17708	Certified-Fresh	98.0	291.0	
17709	Fresh	80.0	10.0	
17710	Fresh	96.0	23.0	
17711	Rotten	50.0	8.0	

	Audience Status	Audience Rating	Audience Count	\
0	Spilled	53.0	254421.0	
1	Upright	64.0	11574.0	
2	Spilled	53.0	14684.0	
3	Upright	97.0	105386.0	
4	Upright	74.0	68918.0	
...	...	...	...	
17707	Upright	74.0	1195.0	
17708	Upright	92.0	101511.0	
17709	Upright	86.0	7146.0	
17710	Upright	91.0	30193.0	
17711	Upright	62.0	4469.0	

	Tomatometer Top Critics Count	Tomatometer Fresh Critics Count	\
0	43	73	
1	44	123	
2	2	16	
3	6	54	
4	5	24	
...	...	...	
17707	2	5	
17708	50	285	
17709	0	8	
17710	6	22	
17711	0	4	

	Tomatometer Rotten Critics Count
0	76
1	19
2	8
3	0
4	3
...	...

17707	4
17708	7
17709	2
17710	1
17711	4

[17106 rows x 22 columns]

```
[45]: # Defining a list of numeric (float64) columns for analysis.
numeric_columns = ["Tomatometer Rating", "Audience Rating", "Tomatometer_
↳Count", "Audience Count"]

# Loop through each column to identify outliers using Z-score
for column in numeric_columns:
    # Calculate the Z-scores
    cleaned_movies_df[f'z_score_{column}'] = stats.
↳zscore(cleaned_movies_df[column])

    # Identify outliers with Z-scores > 3 or < -3
    outliers = cleaned_movies_df[(cleaned_movies_df[f'z_score_{column}'] > 3) |_
↳(cleaned_movies_df[f'z_score_{column}'] < -3)]

    print(f"Outliers in {column}:")
    print(outliers[[column, f'z_score_{column}']])
```

Outliers in Tomatometer Rating:

Empty DataFrame

Columns: [Tomatometer Rating, z_score_Tomatometer Rating]

Index: []

Outliers in Audience Rating:

Empty DataFrame

Columns: [Audience Rating, z_score_Audience Rating]

Index: []

Outliers in Tomatometer Count:

	Tomatometer Count	z_score_Tomatometer Count
169	278.0	3.300460
1306	309.0	3.761389
1623	259.0	3.017955
1693	258.0	3.003086
1837	263.0	3.077430
...	...	...
17560	295.0	3.553227
17600	349.0	4.356136
17628	277.0	3.285591
17684	298.0	3.597833
17708	291.0	3.493752

[332 rows x 2 columns]

Outliers in Audience Count:

	Audience Count	z_score_Audience Count
701	33403994.0	19.981826
1051	32708456.0	19.564103
1310	32104911.0	19.201628
1949	32791717.0	19.614107
2009	32700188.0	19.559137
2046	32167884.0	19.239448
2278	31164074.0	18.636585
2313	33204120.0	19.861786
2662	33781574.0	20.208591
3010	32935203.0	19.700281
4101	33024215.0	19.753740
4447	32495606.0	19.436270
4579	34105114.0	20.402901
4604	31659732.0	18.934265
5211	32749429.0	19.588710
5632	30984568.0	18.528778
6095	32314349.0	19.327412
6453	33356856.0	19.953516
7526	34153607.0	20.432025
7780	32624083.0	19.513430
8335	32604129.0	19.501446
8957	32558003.0	19.473744
9771	31626919.0	18.914558
10183	33324202.0	19.933905
10238	32167706.0	19.239342
10260	33181885.0	19.848433
11251	32601771.0	19.500030
11866	33125722.0	19.814702
12029	32709375.0	19.564655
12544	32459459.0	19.414561
12886	31845749.0	19.045982
12903	32524689.0	19.453737
12923	32582014.0	19.488165
12942	32784354.0	19.609685
13276	34232524.0	20.479420
13336	32222563.0	19.272287
13694	34297354.0	20.518356
13797	33683838.0	20.149893
15410	34679773.0	20.748027
16106	33121539.0	19.812190
17002	31402117.0	18.779548
17035	32518946.0	19.450288
17120	32961772.0	19.716238
17283	32691607.0	19.553984

Tomatometer Rating: No outliers were found, indicating a normal distribution of ratings.

Audience Rating: No outliers were identified, suggesting consistent ratings across the dataset.

Tomatometer Count: Several outliers were detected, ranging from approximately 3.0 to 4.35, indicating that some films received significantly more critic reviews, likely due to popularity or acclaim.

Audience Count: Multiple outliers were found, with extremely high (exceeding 18) review counts for certain films, reflecting strong audience engagement.

```
[46]: # Check for unique values in categorical columns
print(cleaned_movies_df['Genres'].value_counts())
print(cleaned_movies_df['Directors'].value_counts())
```

```
Genres
Drama
1828
Comedy
1233
Comedy, Drama
843
Drama, Mystery & Suspense
702
Art House & International, Drama
573
...
Art House & International, Classics, Cult Movies, Horror, Science Fiction &
Fantasy      1
Action & Adventure, Cult Movies, Drama, Science Fiction & Fantasy
1
Art House & International, Documentary, Sports & Fitness
1
Action & Adventure, Drama, Mystery & Suspense, Special Interest
1
Action & Adventure, Drama, Horror, Kids & Family, Mystery & Suspense
1
Name: count, Length: 1090, dtype: int64
Directors
Unknown Director      189
Woody Allen           36
Clint Eastwood        36
Alfred Hitchcock      34
Sidney Lumet          30
...
Thornton Freeland     1
Giles Borg            1
David Bowers, Sam Fell 1
Carlo Carlei          1
Byron Howard, Rich Moore, Jared Bush 1
Name: count, Length: 8748, dtype: int64
```

Genres: - Drama (1828) and Comedy (1233) dominate, showing a strong audience preference. - Comedy and Drama (843) indicates a blend of these styles. - Genres like Drama, Mystery & Suspense (702) and Art House & International, Drama (573) highlight varied storytelling approaches. - There are 1090 unique genre combinations in the dataset.

Directors: - Unknown Director (189) points to many films lacking prominent names, often from independent projects. - Woody Allen and Clint Eastwood (36 each) are significant figures in the dataset. - Alfred Hitchcock (34) reflects the legacy of classic directors. - The dataset includes 8748 unique directors, showcasing a wide range of talent.

```
[47]: # Check for missing values in the dataset.
missing_values = cleaned_movies_df.isnull().sum()
print("Missing values in each column:")
print(missing_values)

# Check for unusual zero values in important columns.
invalid_runtime = cleaned_movies_df[cleaned_movies_df['Runtime'] == 0]
print("Movies with runtime of 0 (possible invalid data):")
print(invalid_runtime[['Movie Title', 'Runtime']])
```

```
Missing values in each column:
Rotten Tomatoes Link      0
Movie Title               0
Movie Info                0
Critics Consensus         0
Content Rating            0
Genres                   0
Directors                 0
Authors                  0
Actors                   0
Original Release Date     0
Streaming Release Date    0
Runtime                   0
Production Company        0
Tomatometer Status        0
Tomatometer Rating        0
Tomatometer Count         0
Audience Status          0
Audience Rating           0
Audience Count           0
Tomatometer Top Critics Count 0
Tomatometer Fresh Critics Count 0
Tomatometer Rotten Critics Count 0
z_score_Tomatometer Rating 0
z_score_Audience Rating   0
z_score_Tomatometer Count  0
z_score_Audience Count    0
dtype: int64
Movies with runtime of 0 (possible invalid data):
```


	Movie Title	Runtime
170	Night and Day (Bam gua nat)	0.0
217	Prima Ballerina	0.0
273	The Final	0.0
279	Daddy Longlegs	0.0
360	Cat on a Hot Tin Roof	0.0
...	...	...
17501	Wonders Are Many: The Making of Doctor Atomic	0.0
17517	Word Wars	0.0
17542	Wrinkles	0.0
17561	X: Night of Vengeance	0.0
17697	Zombie for Sale	0.0

[302 rows x 2 columns]

No missing values are in any column of the “cleaned_movies_df” DataFrame, indicating complete and reliable data for analysis.

There are 302 movies with a runtime of 0, which is likely an error, as all films should have a defined duration. This raises concerns about data accuracy.

Multiple entries with a runtime of 0 suggest potential data entry issues that require further investigation.

Correct or exclude the zero-runtime entries to ensure data integrity.

4 Human readable dataset after all transformations for “cleaned_critic_reviews_df.”

```
[48]: # Display the cleaned cleaned_critic_reviews_df DataFrame to review the changes
      ↪made during the cleaning process.
      print(cleaned_critic_reviews_df)
```

	Rotten Tomatoes Link	Critic Name	Top Critic	\
0	m/0814255	Andrew L. Urban	False	
1	m/0814255	Louise Keller	False	
2	m/0814255	Unknown Critic	False	
3	m/0814255	Ben McEachen	False	
4	m/0814255	Ethan Alter	True	
...	...	...	...	
1130012	m/zulu_dawn	Chuck O'Leary	False	
1130013	m/zulu_dawn	Ken Hanke	False	
1130014	m/zulu_dawn	Dennis Schwartz	False	
1130015	m/zulu_dawn	Christopher Lloyd	False	
1130016	m/zulu_dawn	Brent McKnight	False	

	Publisher Name	Review Type	Review Score	Review Date	\
0	Urban Cinefile	Fresh	NaN	2010-02-06	
1	Urban Cinefile	Fresh	NaN	2010-02-06	

2	FILMINK (Australia)	Fresh	NaN	2010-02-09
3	Sunday Mail (Australia)	Fresh	3.5/5	2010-02-09
4	Hollywood Reporter	Rotten	NaN	2010-02-10
...	...	...	...	...
1130012	Fantastica Daily	Rotten	2/5	2005-11-02
1130013	Mountain Xpress (Asheville, NC)	Fresh	3.5/5	2007-03-07
1130014	Dennis Schwartz Movie Reviews	Fresh	B+	2010-09-16
1130015	Sarasota Herald-Tribune	Rotten	3.5/5	2011-02-28
1130016	The Last Thing I See	Rotten	C	2020-07-09

	Review Content \
0	A fantasy adventure that fuses Greek mythology...
1	Uma Thurman as Medusa, the gorgon with a coiff...
2	With a top-notch cast and dazzling special eff...
3	Whether audiences will get behind The Lightnin...
4	What's really lacking in The Lightning Thief i...
...	...
1130012	N/A
1130013	Seen today, it's not only a startling indictme...
1130014	A rousing visual spectacle that's a prequel of...
1130015	A simple two-act story: Prelude to war, and th...
1130016	Rides the line between being a pure artifact o...

	Numeric Review Score	z_score
0	3.93438	0.000392
1	3.93438	0.000392
2	3.93438	0.000392
3	3.50000	-0.077095
4	3.93438	0.000392
...	...	...
1130012	2.00000	-0.344670
1130013	3.50000	-0.077095
1130014	3.93438	0.000392
1130015	3.50000	-0.077095
1130016	3.93438	0.000392

[996470 rows x 10 columns]

5 Human readable dataset after all transformations for “cleaned_movies_df.”

```
[49]: # Display the cleaned_movies_df DataFrame to review the changes made during the
      ↪ cleaning process.
      print(cleaned_movies_df)
```

	Rotten Tomatoes Link \
0	m/0814255

1	m/0878835
2	m/10
3	m/1000013-12_angry_men
4	m/1000079-20000_leagues_under_the_sea
...	...
17707	m/zoot_suit
17708	m/zootopia
17709	m/zorba_the_greek
17710	m/zulu
17711	m/zulu_dawn

	Movie Title \
0	Percy Jackson & the Olympians: The Lightning T...
1	Please Give
2	10
3	12 Angry Men (Twelve Angry Men)
4	20,000 Leagues Under The Sea
...	...
17707	Zoot Suit
17708	Zootopia
17709	Zorba the Greek
17710	Zulu
17711	Zulu Dawn

	Movie Info \
0	Always trouble-prone, the life of teenager Per...
1	Kate (Catherine Keener) and her husband Alex (...)
2	A successful, middle-aged Hollywood songwriter...
3	Following the closing arguments in a murder tr...
4	In 1866, Professor Pierre M. Aronnax (Paul Luk...
...	...
17707	Mexican-American gangster Henry Reyna (Daniel ...)
17708	From the largest elephant to the smallest shre...
17709	Traveling to inspect an abandoned mine his fat...
17710	In 1879, the Zulu nation hands colonial Britis...
17711	Sir Henry Bartle Frere's (John Mills) vastly o...

	Critics Consensus	Content Rating \
0	Though it may seem like just another Harry Pot...	PG
1	Nicole Holofcener's newest might seem slight i...	R
2	Blake Edwards' bawdy comedy may not score a pe...	R
3	Sidney Lumet's feature debut is a superbly wri...	NR
4	One of Disney's finest live-action adventures,...	G
...	...	...
17707	No Consensus	R
17708	The brilliantly well-rounded Zootopia offers a...	PG
17709	No Consensus	NR
17710	Zulu patiently establishes a cast of colorful ...	PG

17711

No Consensus

PG

Genres \

0 Action & Adventure, Comedy, Drama, Science Fic...
 1 Comedy
 2 Comedy, Romance
 3 Classics, Drama
 4 Action & Adventure, Drama, Kids & Family
 ...
 17707 Drama, Musical & Performing Arts
 17708 Action & Adventure, Animation, Comedy
 17709 Action & Adventure, Art House & International,...
 17710 Classics, Drama
 17711 Action & Adventure, Art House & International,...

Directors \

0 Chris Columbus
 1 Nicole Holofcener
 2 Blake Edwards
 3 Sidney Lumet
 4 Richard Fleischer
 ...
 17707 Luis Valdez
 17708 Byron Howard, Rich Moore, Jared Bush
 17709 Unknown Director
 17710 Cy Endfield, Cyril Endfield
 17711 Douglas Hickox

Authors \

0 Craig Titley, Chris Columbus, Rick Riordan
 1 Nicole Holofcener
 2 Blake Edwards
 3 Reginald Rose
 4 Earl Felton
 ...
 17707 Luis Valdez
 17708 Jared Bush, Phil Johnston
 17709 Unknown Author
 17710 Cy Endfield, John Prebble
 17711 Cy Endfield, Anthony Storey

Actors \

0 Logan Lerman, Brandon T. Jackson, Alexandra Da...
 1 Catherine Keener, Amanda Peet, Oliver Platt, R...
 2 Dudley Moore, Bo Derek, Julie Andrews, Robert ...
 3 Martin Balsam, John Fiedler, Lee J. Cobb, E.G...
 4 James Mason, Kirk Douglas, Paul Lukas, Peter L...
 ...

17707 Daniel Valdez, Edward James Olmos, Charles Aid...
 17708 J.K. Simmons, Kristen Bell, Octavia Spencer, A...
 17709 Anthony Quinn, Alan Bates, Irene Papas, Lila K...
 17710 Stanley Baker, Jack Hawkins, Ulla Jacobsson, J...
 17711 Burt Lancaster, Peter O'Toole, Simon Ward, Joh...

	Original Release Date	...	Audience Status	Audience Rating	\
0	2010-02-12	...	Spilled	53.0	
1	2010-04-30	...	Upright	64.0	
2	1979-10-05	...	Spilled	53.0	
3	1957-04-13	...	Upright	97.0	
4	1954-01-01	...	Upright	74.0	
...	...	...	...	...	
17707	1981-10-02	...	Upright	74.0	
17708	2016-03-04	...	Upright	92.0	
17709	1964-12-17	...	Upright	86.0	
17710	1964-06-17	...	Upright	91.0	
17711	1979-12-14	...	Upright	62.0	

	Audience Count	Tomatometer	Top Critics Count	\
0	254421.0		43	
1	11574.0		44	
2	14684.0		2	
3	105386.0		6	
4	68918.0		5	
...	...		...	
17707	1195.0		2	
17708	101511.0		50	
17709	7146.0		0	
17710	30193.0		6	
17711	4469.0		0	

	Tomatometer	Fresh Critics Count	Tomatometer	Rotten Critics Count	\
0		73		76	
1		123		19	
2		16		8	
3		54		0	
4		24		3	
...		...		...	
17707		5		4	
17708		285		7	
17709		8		2	
17710		22		1	
17711		4		4	

	z_score_Tomatometer Rating	z_score_Audience Rating	\
0	-0.422902	-0.378467	
1	0.914255	0.161389	

2	0.210488	-0.378467
3	1.371703	1.780959
4	0.984632	0.652168
...	...	...
17707	-0.176583	0.652168
17708	1.301327	1.535569
17709	0.667937	1.241102
17710	1.230950	1.486492
17711	-0.387713	0.063234

	z_score_TomatoMeter Count	z_score_Audience Count
0	1.382400	0.073007
1	1.278319	-0.072841
2	-0.476184	-0.070973
3	-0.030124	-0.016500
4	-0.431578	-0.038402
...	...	...
17707	-0.699215	-0.079074
17708	3.493752	-0.018827
17709	-0.684346	-0.075500
17710	-0.491053	-0.061659
17711	-0.714083	-0.077108

[17106 rows x 26 columns]

6 Ethical implications of data wrangling specific to my data-sources.

In the process of cleaning and transforming the "leaned_critic_reviews_df" and "cleaned_movies"

```
[50]: cleaned_critic_reviews_df.head()
```

```
[50]:  Rotten Tomatoes Link      Critic Name  Top Critic      Publisher Name \
0      m/0814255  Andrew L. Urban      False      Urban Cinefile
1      m/0814255    Louise Keller      False      Urban Cinefile
2      m/0814255   Unknown Critic      False  FILMINK (Australia)
3      m/0814255    Ben McEachen      False  Sunday Mail (Australia)
4      m/0814255    Ethan Alter      True    Hollywood Reporter
```

```
      Review Type  Review Score  Review Date \
0      Fresh      NaN    2010-02-06
1      Fresh      NaN    2010-02-06
2      Fresh      NaN    2010-02-09
3      Fresh    3.5/5    2010-02-09
4      Rotten      NaN    2010-02-10
```

```
      Review Content  Numeric Review Score \
```

0	A fantasy adventure that fuses Greek mythology...	3.93438
1	Uma Thurman as Medusa, the gorgon with a coiff...	3.93438
2	With a top-notch cast and dazzling special eff...	3.93438
3	Whether audiences will get behind The Lightnin...	3.50000
4	What's really lacking in The Lightning Thief i...	3.93438

	z_score
0	0.000392
1	0.000392
2	0.000392
3	-0.077095
4	0.000392

```
[51]: cleaned_movies_df.head()
```

```
[51]:
```

	Rotten Tomatoes Link \
0	m/0814255
1	m/0878835
2	m/10
3	m/1000013-12_angry_men
4	m/1000079-20000_leagues_under_the_sea

	Movie Title \
0	Percy Jackson & the Olympians: The Lightning T...
1	Please Give
2	10
3	12 Angry Men (Twelve Angry Men)
4	20,000 Leagues Under The Sea

	Movie Info \
0	Always trouble-prone, the life of teenager Per...
1	Kate (Catherine Keener) and her husband Alex (...)
2	A successful, middle-aged Hollywood songwriter...
3	Following the closing arguments in a murder tr...
4	In 1866, Professor Pierre M. Aronnax (Paul Luk...

	Critics Consensus	Content Rating \
0	Though it may seem like just another Harry Pot...	PG
1	Nicole Holofcener's newest might seem slight i...	R
2	Blake Edwards' bawdy comedy may not score a pe...	R
3	Sidney Lumet's feature debut is a superbly wri...	NR
4	One of Disney's finest live-action adventures,...	G

	Genres	Directors \
0	Action & Adventure, Comedy, Drama, Science Fic...	Chris Columbus
1	Comedy	Nicole Holofcener
2	Comedy, Romance	Blake Edwards

3	Classics, Drama	Sidney Lumet
4	Action & Adventure, Drama, Kids & Family	Richard Fleischer

	Authors \
0	Craig Titley, Chris Columbus, Rick Riordan
1	Nicole Holofcener
2	Blake Edwards
3	Reginald Rose
4	Earl Felton

	Actors	Original Release Date \
0	Logan Lerman, Brandon T. Jackson, Alexandra Da...	2010-02-12
1	Catherine Keener, Amanda Peet, Oliver Platt, R...	2010-04-30
2	Dudley Moore, Bo Derek, Julie Andrews, Robert ...	1979-10-05
3	Martin Balsam, John Fiedler, Lee J. Cobb, E.G...	1957-04-13
4	James Mason, Kirk Douglas, Paul Lukas, Peter L...	1954-01-01

	... Audience Status	Audience Rating	Audience Count \
0	... Spilled	53.0	254421.0
1	... Upright	64.0	11574.0
2	... Spilled	53.0	14684.0
3	... Upright	97.0	105386.0
4	... Upright	74.0	68918.0

	Tomatometer Top Critics Count	Tomatometer Fresh Critics Count \
0	43	73
1	44	123
2	2	16
3	6	54
4	5	24

	Tomatometer Rotten Critics Count	z_score_Tomatometer Rating \
0	76	-0.422902
1	19	0.914255
2	8	0.210488
3	0	1.371703
4	3	0.984632

	z_score_Audience Rating	z_score_Tomatometer Count	z_score_Audience Count
0	-0.378467	1.382400	0.073007
1	0.161389	1.278319	-0.072841
2	-0.378467	-0.476184	-0.070973
3	1.780959	-0.030124	-0.016500
4	0.652168	-0.431578	-0.038402

[5 rows x 26 columns]


```
[52]: cleaned_critic_reviews_df.to_csv("rt_cleaned_critic_reviews_df.csv",  
    ↪ index=False)
```

```
[53]: cleaned_movies_df.to_csv("rt_cleaned_movies_df.csv", index=False)
```